Can Vision Models Truly Read Charts?
A Multimodal Study of Chart Question Answering

Milan Addepalli
Department of Electrical and Computer Engineering, Rice University
ma200@rice.edu | GitHub Repository

Abstract

Charts convey information through an inherently different visual vocabulary than natural photos (i.e. through heights of bars, slopes of lines, proportions within a pie, etc.). However, most vision encoders are almost exclusively pretrained on natural photographic data, which instead contain real-life objects, such as people, animals, and places. We study whether these encoders can meaningfully understand charts through the task of Chart Question Answering (ChartQA). That is, given a chart image and a natural language question, predict the correct answer. We evaluate three systems on the ChartQA benchmark: a frozen CLIP model plus an MLP classifier, a CLIP model with an LoRA-adapter visual encoder plus an MLP classifier, and zero-shot evaluation with the Qwen2.5-VL-7B. Our results show an exceptionally clear percentage point gap between CLIP-based models agnostic of visual adaptation and SOTA VLMs like Qwen2.5. GradCAM and Attention Rollout independently confirm CLIP attends diffusely rather than localizing to task-relevant features, revealing that the visual domain gap cannot be bridged by lightweight adaptation alone.

1. Introduction

Charts and graphs are some of the most abundant and vital mediums for conveying quantitative information, appearing extensively in academic research, financial analysis, and journalism. The ability to answer natural language questions about these types of figures could be very useful in a variety of fields, yet it remains a surprisingly difficult task for modern vision-language systems.

The difficulty can be traced to a fundamental mismatch, which is that state-of-the-art visual encoders like CLIP [10] are pretrained on hundreds of millions of natural image-text pairs, learning visual representations that are optimized for photographic content. Charts, in contrast, communicate information through an entirely different visual vocabulary. The *height* of a bar, the *slope* of a line, and the *area* of a pie slice bear almost no resemblance to the visual cues that make photo-pretrained models effective. It is therefore far from obvious whether these models can "read" a chart, or whether visual domain adaptation is necessary.

This project investigates that question utilizing the ChartQA benchmark [8]. Chart Question Answering is a chart-specific instance of the broader Visual Question Answering. We evaluate three systems: (1) a **frozen CLIP + MLP** classifier that tests how much chart understanding is already latent in pretrained representations, (2) a **LoRA CLIP + MLP** classifier that applies lightweight visual domain adaptation, and (3) **zero-shot Qwen2.5-VL-7B** [2] as an upper-bound baseline. Beyond accuracy, we perform explainability analysis using GradCAM [11] and Attention Rollout [1] to understand *why* models succeed or fail.

2. Related Work

Chart Understanding. ChartQA [8] provides real-world chart images with human-authored and machine-generated question answer pairs. Prior synthetic benchmarks include FigureQA [5], DVQA [4], and PlotQA [9]. Independently, Liu et al. [7] show that CLIP achieves near-random retrieval on ChartQA and decompose the failure into a vision encoder bottleneck and an extraction bottleneck, finding linear extractors insufficient even when CLIP encodes richer non-linear features, precisely the regime our MLP operates in. MatCha [6] demonstrates that chart-specific pretraining, combining chart derendering and math reasoning objectives, achieves 64.2% on ChartQA, directly validating that domain-specific pretraining at scale can bridge the gap that lightweight adaptation cannot.

Vision Language Models. CLIP [10] is trained through contrastive learning on 400 million image-text pairs. While it transfers well across photographic domains, its contrastive objective aligns global image-text pairs rather than learning the spatial and quantitative structure that charts require, making chart representations a poor fit for down-

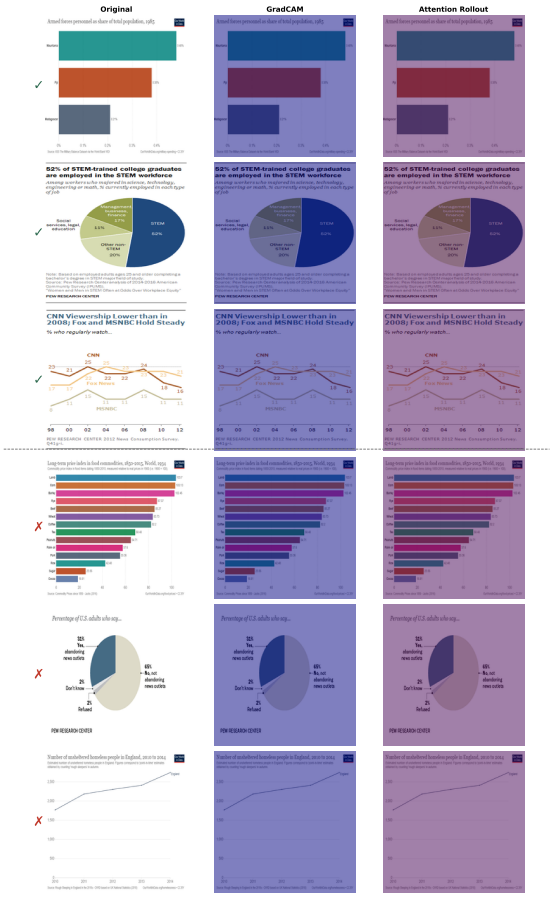

Figure 1. GradCAM and Attention Rollout (LoRA CLIP). **Rows 1–3** (correct): bar chart ("How many bars?" Gold: 3, Pred: 3), pie chart ("Is STEM segment 52%?" Gold: yes, Pred: yes), and line chart ("For how many years was MSNBC viewership above 12?" Gold: 2, Pred: 2). **Rows 4–6** (incorrect): bar chart ("How many food items?" Gold: 14, Pred: 2), pie chart ("What % refused?" Gold: 2, Pred: <UNK>), and line chart ("When does the line reach the peak?" Gold: 2014, Pred: 2). Both GradCAM and Attention Rollout produce identically broad, diffuse heatmaps regardless of correctness, confirming CLIP lacks spatially grounded chart understanding [7].

stream reasoning. Recent general purpose VLMs, such as Qwen2.5-VL [2], are trained on broad multimodal instruction data including documents and figures, enabling strong zero-shot performance on charts. Qwen2.5-VL specifically highlights chart and document understanding as a core training focus, making it the appropriate upper-bound baseline for our experiments.

Parameter-Efficient Fine-Tuning. LoRA [3] reparameterizes weight updates as low-rank matrix products, enabling efficient fine-tuning with a small fraction of trainable parameters. Originally proposed for LLMs, it has since been applied to vision transformers. We apply LoRA to CLIP's visual encoder attention layers to test whether targeted adaptation bridges the chart understanding gap.

Interpretability. GradCAM [11] produces class-discriminative saliency maps from gradients flowing into a target layer. Attention Rollout [1] recursively propagates attention across all transformer layers from the CLS token to each patch. Their agreement provides intersecting evidence.

3. Method

3.1. Architecture

Given a chart image I and question q, the visual encoder f_V and text encoder f_T produce ℓ_2-normalized embeddings $\mathbf{v}, \mathbf{t} \in \mathbb{R}^d$. These are then concatenated and passed through a two-layer MLP to obtain the logits over the answer vocabulary:

$$\hat{y} = W_2 \, \mathrm{ReLU}(W_1[\mathbf{v}; \mathbf{t}] + b_1) + b_2 \tag{1}$$

We use CLIP ViT-B/32 ($d = 512$), MLP hidden dimension on 512, and dropout $p = 0.3$. The answer vocabulary consists of the top-$K = 5{,}000$ most frequent gold answers in the training split, plus a special <UNK> token, giving $K + 1 = 5{,}001$ classes. Test samples whose gold answer falls outside this vocabulary are excluded from accuracy computation, yielding an effective evaluation ceiling of 73.7%.

3.2. Training Variants

Frozen CLIP: both encoders are frozen in this case and only the MLP is trained ($\sim$3.1 million parameters). **LoRA CLIP:** low-rank adapters are applied to the query and value projection matrices of the visual encoder's attention layers. Each target projection W is replaced by:

$$W' = W + \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \; B \in \mathbb{R}^{d_{\text{out}} \times r} \tag{2}$$

with rank $r = 8$ and scaling $\alpha = 16$ using the heuristic $\alpha = 2r$. Only A, B, and the MLP head are updated, adding 294 thousand trainable parameters, which is only 0.34% of the visual encoder.

3.3. Zero-Shot Baseline

Qwen2.5-VL-7B-Instruct [2] is prompted with a short request for a concise answer and decoded greedily with no task-specific training.

3.4. Interpretability Analysis

GradCAM is applied to the last visual encoder block. For predicted class c, the importance weight for each patch position i is the gradient of the class score averaged over the feature dimension:

$$\alpha_i^c = \frac{1}{d} \sum_{j=1}^{d} \frac{\partial y^c}{\partial A_j^i} \tag{3}$$

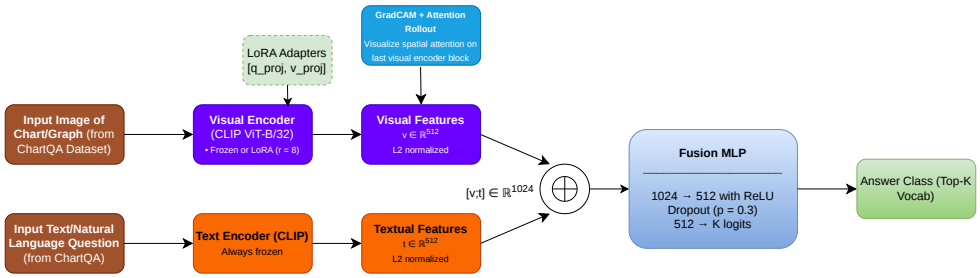

Figure 2. Model architecture. Visual and text encoders are pretrained CLIP components. For the LoRA variant, low-rank adapters are injected into the query and value projection matrices of the visual encoder. The MLP fusion head is always trained from scratch.

The saliency map is then:

$$L^c = \mathrm{ReLU}\left(\sum_i \alpha_i^c A^i\right) \tag{4}$$

where $A^i \in \mathbb{R}^d$ is the activation vector at patch position i. ReLU retains only regions that positively influence the predicted class. Attention Rollout hooks every attention layer, then recursively multiplies attention matrices with residual connections ($A' = 0.5A + 0.5I$) to obtain collective attention starting from the CLS token and through all patches. Both maps are upsampled to image resolution through bilinear interpolation and normalized to $[0, 1]$.

4. Experiments

Dataset. ChartQA [8] contains 28,299 training, 1,920 validation, and 2,500 test samples. Questions are split into *human-authored* and *augmented* types. From the official annotation files, 84% of the charts are bars, 12% are lines, and 4% are pie types.

Evaluation Metric. We report *relaxed accuracy* [8], defined for predicted answer $\hat{y}$ and gold answer y as:

$$\mathrm{correct}(\hat{y}, y) = \begin{cases} 1 & \text{if } \hat{y} = y \\ 1 & \text{if both numeric and } \frac{|\hat{y}-y|}{|y|} \leq 0.05 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

The 5% numerical tolerance accounts for inaccuracies that come from automatic chart data extraction. We further stratify results by question type, chart type, and answer type (binary, numerical, or textual).

Training Details. Both trained models use AdamW as the optimizer with a learning rate of 3×10^{-4} and weight decay of 10^{-4}. They are trained for 20 epochs with a batch size of 64 and cosine annealing for learning rate decay. The best checkpoint is selected by validation accuracy. Both models converge quickly, LoRA CLIP by epoch 5 and Frozen CLIP by epoch 12, with no meaningful improvement thereafter, indicating the bottleneck lies in the visual features rather than training duration. All experiments use either a single NVIDIA Tesla V100 or L40S on Rice University's NOTS cluster.

Table 1. Relaxed accuracy (%) on ChartQA test set by model variant, question type (H=human, A=augmented), and chart type (B=bar, L=line, P=pie.)

Model	Overall	H	A	B	L	P
Frozen CLIP	4.9	7.4	2.4	4.1	8.5	5.2
LoRA CLIP	5.4	7.1	3.6	4.4	9.5	7.0
Qwen2.5-VL	**68.3**	**62.3**	**74.2**	**70.4**	**67.3**	**46.5**

5. Results and Discussion

The gap in chart understanding is profound. The CLIP-based models tested achieve only 4.9-5.4% overall accuracy versus Qwen2.5-VL's 68.3% (Table 1, Figure 3). The scale of this gap suggests CLIP's visual embeddings fundamentally fail to encode chart-discriminative information in a linearly accessible way. The dot products and cosine similarities that a linear classifier relies on carry almost no chart-relevant signal.

LoRA barely helps. LoRA adaptation improves accuracy by only $+0.5$ percentage point (4.9% $\rightarrow$ 5.4%). Adapting 0.34% of the visual encoder's parameters is insufficient to bridge the chart understanding gap. LoRA modifies existing weight directions but cannot reconstruct a chart-specific visual vocabulary never learned during photo-dominated pretraining.

Human vs. augmented reversal. CLIP models perform better on human questions (7.1–7.4%) than augmented ones (2.4–3.6%), while Qwen2.5-VL shows the opposite (62.3% human, 74.2% augmented). Augmented questions require precise numerical answers, which is extremely challenging for a closed-vocabulary classifier but pretty straightforward for a generative model. Human questions involve more common textual answers that appear in the top-5000 vocabulary, explaining CLIP's relative advantage there.

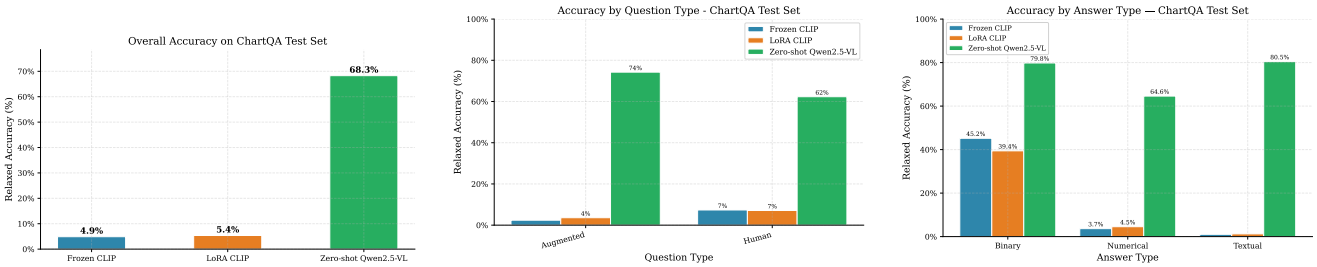

Figure 3. **Left:** Overall relaxed accuracy on the ChartQA test set. The 63 percentage point gap between CLIP models and Qwen2.5-VL is the central finding. **Center:** Accuracy by question type. CLIP performs better on human questions while Qwen2.5-VL shows the opposite pattern, driven by answer type distribution. **Right:** Accuracy by answer type. CLIP predictions collapse to the binary category due to vocabulary frequency bias, with near-zero numerical and textual accuracy. Qwen2.5-VL handles all three meaningfully, with numerical accuracy (80.5%) exceeding binary (79.8%), confirming it genuinely reads chart values.

Answer type stratification. Figure 3 stratifies accuracy by gold answer type. Binary accuracy for CLIP is moderate (39–45%), reflecting that "yes" and "no" are the top-2 vocabulary entries (339 and 255 training occurrences) and the model defaults to them under uncertainty. More surprisingly, textual accuracy is near-zero despite many textual answers appearing in the top-5000 vocabulary. Numerical accuracy is near-zero for both CLIP variants (3.7% and 4.5%), producing the systematic binary predictions for numerical questions visible in Figure 1. In contrast, Qwen2.5-VL achieves 79.8% binary, 80.5% numerical, and 64.6% textual accuracy and notably, its numerical accuracy *exceeds* its binary accuracy, confirming it genuinely reads chart values rather than exploiting answer frequency patterns.

GradCAM and Attention Rollout confirm the perception bottleneck. Figure 1 shows three-column visualizations for both correct (Rows 1–3) and incorrect (Rows 4–6) LoRA CLIP predictions. Both GradCAM and Attention Rollout separately produce broad, diffuse heatmaps with no spatial localization to task-relevant features regardless of correctness. This confirms diffuse attention reflects a fundamental representational limitation rather than a prediction-specific artifact. GradCAM measures gradient flow while Attention Rollout measures structural information flow, so when both independently show diffuse patterns across all cases, CLIP genuinely lacks spatially grounded chart understanding.

6. Conclusion

We investigated whether photo-pretrained vision models can understand charts through the ChartQA task. CLIP-based classifiers achieve only 4.9–5.4% relaxed accuracy regardless of LoRA adaptation, while zero-shot Qwen2.5-VL achieves 68.3%. Both GradCAM and Attention Rollout independently confirm that CLIP attends diffusely across chart regions, providing visual evidence of the linear-regime perception bottleneck independently reported by Liu et al. [7]. Answer type stratification further reveals that CLIP's failures on numerical questions are systematic rather than random, partly exacerbated by the closed-vocabulary classification framing, a framing misaligned with ChartQA's open-ended answer distribution. This is a limitation future work should address by adopting open-vocabulary generation. Bridging this gap requires large-scale multimodal pretraining on document and figure data. Lightweight fine-tuning alone is insufficient.

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